INCH-POUND
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SUPERSEDING
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DETAIL SPECIFICATION

INSULATION, ELECTRICAL, PLASTIC-SEALER COMPOUND

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the plastic-sealer compound used for sealing electrical cable penetrations, end sealing cables, and fire resistant applications requiring electrical insulation with heat and fire-resistant properties.

1.2 Classification. The plastic-sealer compound are of the following types, as specified (see 6.2):

a. Type H – Heat- and fire-resistant

b. Type HF – Heat- and fire-resistant for fire stop applications

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL STANDARDS

FED-STD-313	-	Material Safety Data, Transportation Data and Disposal Data for Hazardous Materials Furnished to Government Activities

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-17	-	Cables, Radio Frequency, Flexible and Semirigid, General Specification for
MIL-PRF-5606	-	Hydraulic Fluid, Petroleum Base; Aircraft, Missile, and Ordnance
MIL-DTL-16884	-	Fuel, Naval Distillate
MIL-PRF-17331	-	Lubricating Oil, Steam Turbine and Gear, Moderate Service

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard, DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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MIL-DTL-24640 - Cables, Lightweight, Low Smoke, Electric for Shipboard Use, General Specification for

MIL-DTL-24643 - Cables, Electric Low Smoke, Halogen-Free, for Shipboard Use General Specification for

MIL-PRF-83282 - Hydraulic Fluid, Fire Resistant, Synthetic Hydrocarbon Base, Metric, NATO Code Number H-537

MIL-PRF-85045 - Cables, Fiber Optic, General Specification for

MIL-PRF-85045/17 - Cable, Fiber Optic, Eight Fibers, Enhanced Performance, Cable Configuration Type 2 (OFCC), Application B (Shipboard), Cable Class SM and MM

MIL-PRF-85045/25 - Cable, Fiber Optic, Seven-Tube, Blown Optical Fiber, Standard and Enhanced Performance, Cable Configuration Type 5 (Tube), Application B (Shipboard), Cable Class SM and MM

MIL-PRF-87257 - Hydraulic Fluid, Fire Resistant; Low Temperature, Synthetic Hydrocarbon Base, Aircraft and Missile

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-129 - Marking for Shipment and Storage

MIL-STD-167-1 - Mechanical Vibrations of Shipboard Equipment (Type I - Environmental and Type II - Internally Excited)

MIL-STD-2003-4 - Electric Plant Installation Standard Methods for Surface Ships and Submarines (Cableways)

MIL-STD-3020 - Fire Resistance of U.S. Naval Surface Ships

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

NAVAL SEA SYSTEMS COMMAND (NAVSEA) DRAWINGS

803-5184182 - Insulation, Passive Fire Protection - Installation Details

(Copies of this document are available online via Model Based Product Support (MBPS) at https://mbps.navsea.cloud.navy.mil/Windchill. To gain access to MBPS, obtain an account with National Help Desk Service Management (NHDSM) at https://nhdsm.navair.navy.mil (a valid CAC is required to access this website) and submit a SAAR-N Request. Refer questions, inquiries, or problems to (888) 292-5919.)

NAVAL SEA SYSTEMS COMMAND (NAVSEA) PUBLICATIONS

T9070-AL-DPC-020/077-2 - NAVSEA Hazardous Material Avoidance Process

(Copies of this document are available online via Model Based Product Support (MBPS) at https://mbps.navseaplm.navy.mil/Windchill/app/. To gain access to MBPS, obtain an account with National Help Desk Service Management (NHDSM) at https://nhdsm.navair.navy.mil (a valid CAC is required to access this website) and submit a SAAR-N Request. Refer questions, inquiries, or problems to (888) 292-5919. This document is available for ordering (hard copy) via the Naval Logistics Library (NLL) at https://nll.navsup.navy.mil. For questions regarding the NLL, contact the NLL Customer Service at nllhelpdesk@navy.mil, (866) 817-3130, or (215) 697-2626/DSN 442-2626.)

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2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI Z400.1/Z129.1	-	Hazardous Workplace Chemicals - Hazard Evaluation and Safety Data Sheet and Precautionary Labeling Preparation

(Copies of this document are available online at https://webstore.ansi.org/.)

ASTM INTERNATIONAL

ASTM D5/D5M	-	Standard Test Method for Penetration of Bituminous Materials
ASTM D149	-	Standard Test Method for Dielectric Breakdown Voltage and Dielectric Strength of Solid Electrical Insulating Materials at Commercial Power Frequencies
ASTM D257	-	Standard Test Methods for DC Resistance or Conductance of Insulating Materials
ASTM D412	-	Standard Test Methods for Vulcanized Rubber and Thermoplastic Elastomers—Tension
ASTM D635	-	Standard Test Method for Rate of Burning and/or Extent and Time of Burning of Plastics in a Horizontal Position
ASTM D770	-	Standard Specification for Isopropyl Alcohol
ASTM D2240	-	Standard Test Method for Rubber Property—Durometer Hardness
ASTM D3487	-	Standard Specification for Mineral Insulating Oil Used in Electrical Apparatus
ASTM E28	-	Standard Test Methods for Softening Point of Resins Derived from Pine Chemicals and Hydrocarbons, by Ring-and-Ball Apparatus

(Copies of these documents are available online at www.astm.org.)

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO/IEC 17025	-	General Requirements for the Competence of Testing and Calibration Laboratories

(Copies of this document are available online at www.iso.org/home.html.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. The plastic-sealer insulation compound furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.3 and 6.3).

3.2 Materials. The materials used shall have the physical and electrical properties as specified herein. The materials shall not, either in the state of their original application or as a result of aging, have any deleterious effect on insulation materials to which they are applied.

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3.2.1 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.2.2 Toxicity. When evaluated in accordance with 4.5.2.1, the plastic-sealer insulation compound shall pose no serious or high risk to the health of personnel or the environment when used for its intended purpose (see 4.5.2.1 and 6.8).

3.2.3 Prohibited materials. The plastic-sealer insulation compound shall not contain any chemicals categorized as "prohibited" in accordance with T9070-AL-DPC-020/077-2.

3.3 Application characteristics. The plastic-sealer insulation compound, in the condition in which delivered, shall be effective without the addition of other ingredients. The consistency at ambient room temperature shall be such that it can be readily applied by means of the fingers or with a knife, and shall be capable of being drawn or kneaded into long thin rolls. The temperature required for application of the compound shall not exceed 86 degrees Fahrenheit (°F) (see 4.5.3).

3.4 Hardening properties. When measured by the method specified in 4.5.4, the penetration to 0.004 inch shall be as follows:

$$\frac{\text{Minimum penetration at } 32\ ^\circ\text{F}}{5} \qquad \frac{\text{Maximum penetration at } 77\ ^\circ\text{F}}{100}$$

3.5 Aging. After aging, the compound shall have a minimum penetration of 25 (0.004 inch) when tested as specified in 4.5.5.

3.6 Softening point. The softening point of the compound, after it is fully hardened, shall be not less than 212 °F when tested as specified in 4.5.6.

3.7 Adhesiveness. The compound, when applied to the surfaces of two flat steel discs as specified in 4.5.7, shall have an adhesiveness that requires not less than 100 pounds for separation of the plates.

3.8 Fluid-proofness. The fluid-proofness of the compound shall be such that the change in weight of sample shall be not more than 10 percent when tested as specified in 4.5.8.

3.9 Waterproofness. The waterproofness of the compound shall be such that the increase in weight of a sample shall be not greater than 0.5 percent when tested as specified in 4.5.9.

3.10 Insulation resistance. The compound shall have an insulation resistance of not less than 50 megohms when measured as specified in 4.5.10.

3.11 Dielectric strength. A test specimen prepared as specified in 4.5.10 shall show no evidence of breakdown when tested as specified in 4.5.11.

3.12 Compatibility. The compound shall not cause any significant deterioration in the characteristics of the cable electrical jackets listed in table I when tested as specified in 4.5.12. In order to minimize the number of cable test specimens for compatibility tests, only the cable materials as designated for tests in table I need to be tested. The compound being tested shall be considered incompatible if one or more of the following is evident in the heat-aged, treated specimens when they are compared to the heat-aged, untreated specimens as specified in 4.5.12.2:

a. The heat-aged, treated specimens exhibit signs of swelling, softening, hardening, cracking, or other visible deterioration that is not evident in the heat-aged, untreated specimens.

b. The shore A hardness results vary by more than ±10 percent.

c. The elongation results vary by more than ±15 percent.

d. The reduction in tensile strength of the heat-aged, treated specimens is greater than 15 percent.

e. The reduction in insulation resistance of the heat-aged, treated specimens is greater than 10 percent.

f. There is reduction in the dielectric strength of the heat-aged, treated specimens.

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TABLE I. Cable jacket materials.

Defense specification	Material
MIL-DTL-24643 (see 6.6)	Crosslinked Polyolefin (XLPO) 1/ 2/
MIL-DTL-17 (see 6.6)	Crosslinked Polyolefin (XLPO) Type XIV 1/ 2/
MIL-PRF-85045 (see 6.6)	Jacket material shall be composed of a low-halogen, low-smoke, low-toxicity polymer material. 1/ 2/
MIL-DTL-24640 (see 6.6)	Crosslinked Polyolefin (XLPO) 1/ 2/
NOTES: 1/ Test required. 2/ Jacket material.	

3.13 Flammability. Type H and HF materials shall be deemed as self-extinguishing when tested as specified in 4.5.13.

3.14 Vibration, shock, and fire resistance test requirements for type HF material.

3.14.1 Vibration requirements. Type HF material shall not allow passage of pressurized air before or after a sample cableway penetration assembly is tested as specified in 4.5.14.2.1.

3.14.2 Shock test requirements. Type HF material shall not allow passage of pressurized air after a sample cableway penetration assembly is tested as specified in 4.5.14.2.2.

3.14.3 Fire resistance test requirements. Type HF material when tested as specified in 4.5.14.2.3 shall prevent the passage of fire and hot gases to the unexposed side of the penetration and shall limit the temperature rise per the requirements of MIL-STD-3020 fire resistance criteria for Class N-30 divisions.

3.14.4 Hose stream test requirements. When a hose stream test is conducted in accordance with MIL-STD-3020 and as specified in 4.5.14.2.4, no openings shall develop during the application of the stream that allow water to pass to the unexposed face.

3.15 Expansion. Type HF material shall have an expansion volume of 5 percent minimum when tested as specified in 4.5.15.

3.16 Marking.

3.16.1 Instructions for use. Each can or pail of sealant shall have attached thereto a label describing the method of application.

3.16.2 Warnings. Each can or pail shall have attached thereto appropriate warning labels in accordance with MIL-STD-129 and as specified (see 6.2).

3.17 Workmanship. All compounds of the same brand, from one manufacturer, shall be the same color and uniform in texture and shall be homogenous.

3.18 Safety data sheet. When specified (see 6.2), a completed safety data sheet (SDS) shall be provided in accordance with ANSI Z400.1/Z129.1 or FED-STD-313.

4. VERIFICATION

4.1 Classification of inspections. The inspections specified herein are classified as follows:

a. Qualification inspection (see 4.3).

b. Conformance inspection (see 4.4).

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4.2 Inspection conditions. Unless otherwise specified herein, all inspections shall be conducted at a temperature of 73.4±5 °F and a relative humidity of 50±5 percent. The temperature of the material as stored in the open container shall be at room temperature for a period of 24 hours before being tested. Material for test shall not be removed from the open container by hand, but by a suitable instrument.

4.2.1 Fire testing provisions. All fire tests specified in this document shall be conducted by an independent testing laboratory that is accredited to ISO/IEC 17025 and is acceptable to NAVSEA. Accreditation shall be obtained from a recognized accreditation body such as the American Association for Laboratory Accreditation (A2LA) (www.a2la.org) or the International Code Council's International Accreditation Services (IAS) (www.iasonline.org). The scope of accreditation shall include specific flammability and fire tests required for qualification. All other fire test provisions shall be as specified (see 6.2 and 6.9).

4.3 Qualification inspection. Qualification inspection shall be conducted at a laboratory acceptable to NAVSEA. Qualification inspection shall consist of the examination and tests specified in table II.

TABLE II. Qualification inspections.

Inspection	Requirement	Test method
Visual examination	3.16 and 3.17	4.5.1
Toxicity	3.2.2	4.5.2.1
Prohibited materials	3.2.3	4.5.2.2
Application properties	3.3	4.5.3
Hardening properties	3.4	4.5.4
Aging	3.5	4.5.5
Softening point	3.6	4.5.6
Adhesiveness	3.7	4.5.7
Fluid-proofness	3.8	4.5.8
Waterproofness	3.9	4.5.9
Insulation resistance	3.10	4.5.10
Dielectric strength	3.11	4.5.11
Compatibility	3.12	4.5.12 through 4.5.12.2.7
Flammability	3.13	4.5.13
Soap and blow 1/	3.14.1	4.5.14.2.1
Vibration 1/	3.14.1	4.5.14.2.1
Soap and blow 1/	3.14.1	4.5.14.2.1
Shock 1/	3.14.2	4.5.14.2.2
Soap and blow 1/	3.14.2	4.5.14.2.2
Fire resistance 1/	3.14.3	4.5.14.2.3
Hose stream 1/	3.14.4	4.5.14.2.4
Expansion 1/	3.15	4.5.15

NOTE:
1/ Type HF only.

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4.3.1 Sample size. A 5-gallon sample of plastic-sealer insulation compound shall be prepared to complete all required inspections and tests.

4.3.2 Failure. Failure to meet the requirements specified herein shall be cause for disapproval of or removal of product qualification.

4.3.3 Periodic retention of qualification. At 5-year intervals after initial qualification, inspection for retention of qualification shall be performed on plastic-sealer insulation compound and shall include the inspections specified in table III. The inspection shall be performed on one sample of each type of qualified plastic-sealer insulation compound.

4.3.4 Change approval. Changes in construction, formulation, materials, design, or manufacturing process of qualified products may require requalification. Unless otherwise specified by the qualifying activity, changes require written approval by NAVSEA.

4.4 Conformance inspection. Conformance inspection shall consist of the examination and tests specified in table III.

TABLE III. Conformance and retention inspections.

Inspection	Requirement	Test method
Visual examination	3.16 and 3.17	4.5.1
Application properties	3.3	4.5.3
Hardening properties	3.4	4.5.4
Aging	3.5	4.5.5
Waterproofness	3.9	4.5.9
Insulation resistance	3.10	4.5.10
Dielectric strength	3.11	4.5.11
Flammability	3.13	4.5.13

4.4.1 Inspection batch. An inspection batch (see 6.5.1) shall consist of all plastic-sealer insulation compound of the same type designation, produced under essentially the same conditions, and offered for inspection at one time.

4.4.2 Sampling plan. A 1-pound sample selected from each batch of compound shall be subjected to the conformance examination and tests in table III, to determine conformance to this specification. Failure to conform to the requirements of this specification for any test shall be counted as a defect and the batch shall be rejected.

4.5 Methods of inspection.

4.5.1 Visual examination. The sample plastic-sealer compound shall be examined to verify compliance with the marking requirements of 3.16 and workmanship requirements of 3.17.

4.5.2 Materials.

4.5.2.1 Toxicity. A Health Hazard Assessment (HHA) will be conducted to ensure conformance to 3.2.2 as required by the qualifying activity. The Navy and Marine Corps Force Health Protection Command (NMCFHPC) will evaluate the plastic-sealer insulation compound using data provided by the manufacturer/distributor to the NMCFHPC (see 3.2.2 and 6.8).

4.5.2.2 Prohibited materials. Prohibited materials shall be verified for conformance to 3.2.3 as required by the qualifying activity.

4.5.3 Application properties. The compound, when applied in accordance with the manufacturer's instructions (see 3.16.1), shall meet the requirements of 3.3.

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4.5.4 Hardening properties. The hardening properties of the compound shall be within the values specified in 3.4, when tested in accordance with ASTM D5/D5M. The following details shall apply:

a. Preparation of sample – maintained at room temperature for 72 hours after packing into the container.

b. Water bath temperature – 32 or 77 °F.

c. Needle – loaded with 1.76 ounces (50 grams [g]).

d. Time interval – 5 seconds.

e. Number of readings – not less than three, whose values differ not more than 0.016 inch.

4.5.5 Aging. The compound, after aging for 96 hours at 158 °F, shall be subjected to the test in 4.5.4 at a temperature of 77 °F. The minimum penetration shall be as specified in 3.5.

4.5.6 Softening point. The softening properties of the compound shall be as specified in 3.6, when tested in accordance with ASTM E28. The following details and exceptions shall apply:

a. Preparation of sample – allowed to harden at room temperature for 72 hours prior to the test.

b. Distilled water cooling – not less than 81 °F below subtracting point of material under test.

c. Stirrer – use optional. If not used, uniform heating may be achieved by placing the burner midway between the center and edge of the beaker, away from the specimen.

4.5.7 Adhesiveness. The compound shall be applied to the surface of two flat steel disks 4 inches in diameter. The surfaces of the steel disks shall be finished to a standard surface roughness of 8±1 micro-inches, root mean square (rms). The disks shall be pressed together in a vise until the thickness of the sample is 1/16 inch. All excess material shall be removed and the compound allowed to harden for 72 hours, after which the pull-apart stress shall be determined in a tensile testing machine. A strain rate of 1 inch per minute shall be used for the stress test. The material shall meet the adhesiveness requirements specified in 3.7.

4.5.8 Fluidproofness. Three corrosion-resistant steel AISI type 302/304 cylindrical tubes, 1 inch in diameter and ¾ inch long, shall be cleaned with a solvent to remove all traces of dirt and grease. The clean tubes shall then be weighed on an analytical balance to the nearest 0.0001 g (0.00000353 ounces). A spatula shall be used to pack the tubes full of sealer compound, to ensure the absence of air pockets. The compound shall be leveled flush with the ends of the tube and all excess compound removed from the outer surface of the tube. The filled tubes shall be weighed on an analytical balance and then immersed in transformer oil, in accordance with ASTM D3487, which shall be held at a temperature of 104 °F for 48 hours. Both surfaces of the compound shall be exposed to the transformer oil by laying the tubes on their sides. After removal from the transformer oil bath, the excess transformer oil on the surface of the compound shall be blown off with a soft stream of air, and the metal surfaces dried by means of a paper towel. The filled tubes shall be reweighed on an analytical balance. The average percent increase or decrease in net weight shall then be calculated from readings obtained on three specimens. If the change in weight is greater than 10 percent, the sample shall be considered as having failed the test (see 3.8).

a. Repeat 4.5.8 test, replacing the transformer oil with isopropyl alcohol in accordance with ASTM D770. The test shall be conducted at room temperature (68 to 77 °F) for 12 hours.

b. Repeat 4.5.8 test, replacing the transformer oil with hydraulic oil in accordance with MIL-PRF-5606. The test shall be conducted at 104 °F for 24 hours. Note: MIL-PRF-83282 and MIL-PRF-87257 are acceptable alternative fluids for use in place of MIL-PRF-5606, at the manufacturer's option.

c. Repeat 4.5.8 test, replacing the transformer oil with Coolant Monsanto Coolanol 25 or equal. The test shall be conducted at room temperature (68 to 77 °F) for 24 hours.

d Repeat 4.5.8 test, replacing the transformer oil with lubricating oil in accordance with 2190 TEP, MIL-PRF-17331. The test shall be conducted at 104 °F for 12 hours.

e. Repeat 4.5.8 test, replacing the transformer oil with fuel oil in accordance with MIL-DTL-16884. The test shall be conducted at 104 °F for 24 hours.

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4.5.9 Waterproofness. The compound shall be packed in corrosion-resistant steel cylindrical tubes and weighed as specified in 4.5.8. The filled tubes shall be immersed in distilled water at room temperature for 24 hours. After removal from the water, the excess water shall be removed as specified in 4.5.8, and the tubes shall be reweighed. The average percent increase in net weight shall then be calculated from readings obtained on three specimens. If the increase in weight is greater than 0.5 percent, the sample shall be considered as having failed the test (see 3.9).

4.5.10 Insulation resistance. Insulation resistance shall be measured with a 1,000-volt (V) insulation resistance indicating instrument using a 1¼-inch diameter electrode, placed centrally on the top side of a 2-inch diameter by ½-inch thick disk of the compound. The test material shall be held in place by insulating tubing. The assembly shall rest on a 3-inch square brass plate. Failure to meet the requirements of 3.10 shall be cause for rejection.

4.5.11 Dielectric strength. The test specimen, prepared in accordance with 4.5.10, shall successfully withstand an alternating current (AC) potential of 1,500 Vrms, applied instantaneously and maintained for not less than 1 minute, using a ¾-inch diameter electrode located externally at the top of the specimen. The test voltage shall approximate a true sine wave, and shall be of a frequency not exceeding 100 hertz. Failure to meet the requirements of 3.11 shall be cause for rejection.

4.5.12 Compatibility. The compatibility of the compound with each type of cable jacket material shall be determined as specified in table I (see 3.12).

4.5.12.1 Specimens. Nine 6-inch specimens of each type material in table I are required for the compatibility tests. Each group of nine specimens shall be taken from a single length of cable.

4.5.12.1.1 Preparation of specimens. The specimens shall be stripped from the cable and shall be cleaned, using a solvent that is compatible with the cable jacket material to ensure the cable is free from any contaminants. Cleaning of specimens is necessary to ensure repeatability of test results. Three of the nine specimens in each group shall be either packed with or compressed between layers of the sealer compound. The remaining six specimens in each group shall be left untreated.

4.5.12.2 Procedure. The three specimens treated with the compound and three of the untreated specimens shall be heat-aged; the remaining three untreated specimens of the group shall be kept in the “as received” condition. After heat-aging, testing shall be performed on the three treated and six untreated specimens in each group at the same time. The sealing compound shall be stripped from the treated specimens before tests are performed. The average mean of the results of each test shall be calculated for the specimens in each set of three, then the mean of the treated specimens shall be compared with the mean of each set of the untreated specimens. The compound shall be considered as failing the test if any of the observations specified in 3.12 are evident.

4.5.12.2.1 Heat-aging. The six prepared specimens (see 4.5.12.1.1) in each group shall be subjected to heat-aging for 168 hours at a temperature of 160 °F. The aging apparatus shall consist of a heat chamber, a means of controlling the temperature, a thermometer or thermocouple for measuring temperature, and racks or other means of supporting the specimens within the chamber. The temperature control shall be capable of maintaining the chamber temperature at 160±5 °F.

4.5.12.2.2 Visual examination. The heat-aged specimens shall be subjected to a thorough examination for signs of deterioration as specified in 3.12.

4.5.12.2.3 Dielectric strength. Each group of specimens shall be subjected to dielectric strength testing in accordance with ASTM D149 to the average dielectric strength of the heat-aged, untreated specimens, as specified in 3.12. The following details and exceptions shall apply:

a. The test medium shall be the normal laboratory atmospheric conditions.

b. The electrodes shall be ¼-inch diameter.

c. The applied voltage shall be 500 V AC (Vrms).

d. Using a suitable measuring device, the leakage current shall be measured after 1 minute of electrification.

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4.5.12.2.4 Insulation resistance. Each group of specimens shall be subjected to resistance testing in accordance with ASTM D257. The average insulation resistance of the treated specimens when compared to the average insulation resistance of the heat-aged, untreated specimens shall be as specified in 3.12. The following details and exceptions shall apply:

a. The test medium shall be the normal laboratory atmospheric conditions.

b. The applied voltage shall be 100 V direct current.

c. The measured resistance shall be specified in ohms.

4.5.12.2.5 Indentation hardness. Each group of specimens shall be subject to shore A hardness determination in accordance with ASTM D2240. The average indentation hardness of the treated specimens when compared to the average indentation hardness of the heat-aged, untreated specimens shall be as specified in 3.12.

4.5.12.2.6 Tensile strength. Each group of specimens shall be subject to tensile strength tests in accordance with ASTM D412. The average tensile strength of the treated specimens when compared to the average tensile strength of the heat-aged, untreated specimens shall be as specified in 3.12.

4.5.12.2.7 Elongation. Each group of specimens shall be subject to elongation tests in accordance with ASTM D412. The bench marks shall be 2 inches apart. The average ultimate elongation of the treated specimens when compared to average ultimate elongation of the heat-aged, untreated specimens shall be as specified in 3.12.

4.5.13 Flammability. A sample shall be prepared by molding the sealant into a 5-inch long by 0.5-inch wide by 0.25-inch thick shape. Once prepared, the sample shall be tested in accordance with ASTM D635. Acceptability criteria shall be as specified in 3.13.

4.5.14 Vibration, shock, fire resistance, and hose stream test requirements. The electrical insulation plastic-sealer compound shall meet:

a. Vibration requirements in accordance with MIL-STD-167-1.

b. The shock, fire resistance, and hose stream test requirements in accordance with MIL-STD-3020 appendices for:

1. “Method of Fire Tests for Fire Resistance of Bulkheads and Decks (Overheads);”
2. “Hose Stream Test Configuration;”
3. “Cable Transit Test Configuration;” and
4. “Shock Test Prior to Fire Resistance Test Configuration.”

See MIL-STD-3020 for the applicability of these appendices. When specified by the qualifying activity, detailed test procedures for vibration, shock, fire resistance, and hose stream require written approval by NAVSEA.

4.5.14.1 Test equipment. The test equipment for conducting the vibration, shock, fire stop resistance, and hose stream tests shall be as follows:

a. Simulated steel bulkhead and deck assemblies for test shall be built in accordance with “Method of Fire Tests for Fire Resistance of Bulkheads and Decks” appendix of MIL-STD-3020, except that the bulkhead and deck assemblies shall be 4 by 6 feet (1,218 by 1,829 mm) for all tests. Collars, as shown on figure 1, shall be 8-inch diameter Schedule 40 steel pipe 3-inch length (minimum).

b. Sixteen 120-inch lengths of 13 different types of cables in accordance with MIL-DTL-17, MIL-DTL-24643, MIL-DTL-24640, and MIL-PRF-85045, as applicable (see figure 1).

c. MIL-PRF-85045/25 fiber optic cable (shown on figure 1) tube ends shall be plugged with a non-flammable sealant to simulate end caps. The tube end sealant is used to make non-open tube systems for the fire resistance test. Shipboard end caps used internal to equipment are not used for this test based on the likely proximity of endcaps’ direct exposure to fire. Sealing of the open tube systems is essential to the setup for test execution and the consistent manner in which MIL-PRF-85045/17 and MIL-PRF-85045/25 are tested for qualification.

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4.5.14.2 Test procedure. The required lengths of cable shall be mounted in the collars as shown on figure 1. The cables shall be supported in accordance with the requirements of "Cable Transit Test Configuration" appendix of MIL-STD-3020 and "Hanger loading, spacing, and tier limitations" of MIL-STD-2003-4. The plastic-sealer compound shall be packed inside the penetration around the cables to a depth of 3 inches.

4.5.14.2.1 Vibration test. Fire resistant divisions and their penetrations shall be subjected to a vibration test in accordance with MIL-STD-167-1 prior to conducting the shock test of MIL-STD-3020 with the exception that fire insulation materials and attachment methods are not evaluated as part of this test. The specific shipboard mounting and location shall be determined and adequately represented in the vibration test. The test specimen construction, orientations, fixture, and assembly details are provided for guidance in the "Shock Test Prior to Fire Resistance Test Configuration" appendix of MIL-STD-3020. Qualification testing shall be performed using both deck (overhead) and bulkhead orientations using the collars shown on figure 1.

a. The cable and plastic-sealer compound shall not demonstrate loosening or degradation that may affect fire performance. The penetration integrity shall be verified before and after the vibration test by an air hose nozzle and soapy water test. This test is performed using an air hose nozzle that shall be held 3 inches from the sealant surface. The distance shall be not closer than 3 inches to avoid distortion of the sealant due to high air flow. The minimum nozzle diameter shall be ⅜ inch and the nozzle pressure shall be 30 to no more than 60 pounds per square inch (207 to 414 kilopascal). A soapy solution shall be applied to the opposite face of the cable collar. The air pressure shall then be directed by moving the nozzle back and forth in an even manner across the entire surface of the sealant. The soapy film on the opposite face of the cable collar shall be inspected for leaks evidenced by air bubbles forming on the soapy film. If there are no bubbles on the other side, the vibration-tested assembly shall then be subjected to the shock test. If bubbles are present on the other side of the assembly, air has been allowed to pass through the assembly and it has failed the vibration test as specified in 3.14.1. Retesting to meet vibration requirements is at the discretion of the qualifying activity.

b. When specified by the qualifying activity, previous or related vibration test results and analysis to validate fire resistance is not degraded by vibration may be acceptable.

4.5.14.2.2 Shock test. The shock test shall be conducted in accordance with "Shock Test Prior to Fire Resistance Test Configuration" appendix of MIL-STD-3020, with the exception that fire insulation materials and attachment methods are not evaluated as part of this test.

a. The cable and plastic-sealer compound shall not demonstrate loosening or degradation that may affect fire performance. The penetration integrity shall be verified before and after the shock test by an air hose nozzle and soapy water test performed using the method in 4.5.14.2.1.a. The soapy film on the opposite face of the cable collar shall be inspected for leaks evidenced by air bubbles forming on the soapy film. If there are no bubbles on the other side, the shock-tested assembly shall then be subjected to the fire resistance test. If bubbles are present on the other side of the assembly, air has been allowed to pass through the assembly and it has failed the shock test as specified in 3.14.2. Retesting to meet shock requirements is at the discretion of the qualifying activity.

b. When specified by the qualifying activity, previous or related shock test results and analysis to validate fire resistance is not degraded by shock may be acceptable.

4.5.14.2.3 Fire resistance test. The cable and plastic-sealer compound should be visually inspected prior to fire resistance testing. The fire resistance test shall be conducted in accordance with "Method of Fire Tests for Fire Resistance of Bulkheads and Decks" and "Cable Transit Test Configuration" appendices of MIL-STD-3020. Instrumentation shall be positioned in accordance with "Cable Transit Test Configuration" appendix of MIL-STD-3020. Deck and bulkhead insulation shall be in accordance with the restricted configuration of 803-5184182. Failure to meet the requirements of 3.14.3 shall be cause for rejection. Retesting to meet fire resistance requirements is at the discretion of the qualifying activity.

4.5.14.2.4 Hose stream test. The hose stream test shall be conducted in accordance with "Hose Stream Test Configuration" appendix of MIL-STD-3020. The hose stream test shall be deemed acceptable in accordance with "Cable Transit Test Configuration" appendix of MIL-STD-3020. Failure to meet the requirements of 3.14.4 shall be cause for rejection.

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4.5.15 Expansion. The compound shall be packed inside the tube section (see figure 2) to a depth of approximately 3 inches. The tube shall then be mounted into the tube holder (see figure 3). A measuring washer shall be placed on top of the material inside the tube with the indentation pointing up. The distance from the washer indent to the top surface of the tube shall be measured with a depth gage. The total assembly shall then be placed into an oven with the temperature held at 608 °F for 1 hour. After the sample has cooled to room temperature, the depth from the top surface of the tube to the washer indent shall be measured again. The change in volume can be found by:

$$\text{percent (expansion)} = \frac{D_1 - D_2}{6 - D_1}$$

where D1 = depth before heating

D2 = depth after heating

The compound shall be considered as having failed the test if the expansion in volume is less than 5 percent.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The compound will be used for sealing electrical cable penetrations in decks and bulkheads and entrances in electrical enclosures and will also be used for end sealing cables. The type HF compound is also intended for use as a fire stop at cable penetrations in decks and bulkheads. The compound will be in direct contact with the exterior surfaces of both armored and unarmored cables and conductor insulation materials. Current stock of compound may be used until depleted; future requisitions for compound should refer to the current version of this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Type and quantity of compound required (see 1.2).

c. When a safety data sheet (SDS) is required (see 3.18 and 6.4).

d. Additional fire testing provisions (see 4.2.1 and 6.9).

e. Packaging requirements (see 5.1).

f. Requirements for certified test reports (see 6.2.1).

6.2.1 Certified test reports. The contractor should prepare test reports, including those for visual examination, toxicity, application properties, hardening properties, aging, softening point, adhesiveness, fluid-proofness, waterproofness, insulation resistance, dielectric strength, compatibility, flammability, vibration (including soap and blow), shock (including soap and blow), fire resistance including hose stream, and expansion as specified (see 6.2). The test report should contain the following information:

a. Serial number and date of NAVSEA letter of approval.

b. A statement to the effect that the product was constructed from materials listed on the manufacturer's approved details of construction sheet.

c. A statement that the product meets all of the requirements of this specification.

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d. Results of all conformance tests showing actual values obtained.

e. Year and month the compound was manufactured.

f. Customer's name and contract or order number.

g. A statement that the product is free of prohibited materials.

h. The serial number of all NAVSEA letters approving deviations.

i. Name and address of the test laboratory that conducted the tests and test report numbers when applicable.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List No. QPL-3064 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://assist.dla.mil.

6.3.1 Provisions governing qualification. Copies of SD-6, "Provisions Governing Qualification," are available online at https://quicksearch.dla.mil.

6.4 Safety data sheet (SDS). When specified (see 6.2), contracting officers will identify those activities requiring copies of completed SDSs prepared in accordance with Appendix D of 29 CFR 1910.1200. In order to obtain the SDS, federal acquisition regulation (FAR) clause 52.223-3 will be in the contract. The contracting activity should be given an SDS at the time of contract award.

6.5 Definitions.

6.5.1 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process and subjected to some physical mixing operation intended to make the final product substantially uniform.

6.6 Cables for compatibility test. The following specification sheets list cables that have cable jacket materials listed in table I for which tests are required. This information is given for the convenience of the plastic-sealed compound manufacturer in selecting cables for compatibility testing of the compound. Since this list is not all inclusive, the manufacturer may select other cables listed in MIL-DTL-17, MIL-DTL-24640, MIL-DTL-24643, and MIL-PRF-85045 provided they have the required cable jacket material. The qualified manufacturers of the cables can be found in the applicable QPL.

a. Crosslinked Polyolefin (XLPO) jacket (MIL-DTL-24643/14, /16, /17, /22, /23, /28, /30, /33, and /46).

b. Crosslinked Polyolefin (XLPO) jacket (MIL-DTL-17/189, /190, and /191).

c. Crosslinked Polyolefin jacket (MIL-DTL-24640/7, /14, /19, and /22).

d. Jacket material composed of a low-halogen, low-smoke, low-toxicity polymer material (MIL-PRF-85045/17 and /25).

6.7 Sub-contracted material and parts. The preparation for delivery requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.8 Toxicity evaluation. The NMCFHPC requires sufficient information to permit an HHA of the product. Upon completion of the HHA, a copy will be provided by the NMCFHPC to the Government for evaluation. The HHA process is described on the NMCFHPC's website, https://www.med.navy.mil/Navy-and-Marine-Corps-Force-Health-Protection-Command/Environmental-Health/Industrial-Hygiene/Acquisition-Technical-Support/Health-Hazard-Assessments-HHA/.

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6.9 Additional fire testing provisions. NAVSEA reserves the right to witness the tests and perform any of the tests set forth herein where such testing is deemed necessary to assure compliance to prescribed requirements of the qualification tests. A2LA (www.a2la.org) or International Code Council's IAS (www.iasonline.org) maintain a full list of the approved laboratories that are accredited to ISO/IEC 17025 and provide details on how to obtain and maintain accreditation (see 6.2).

6.10 Subject term (key word) listing.

Cable end sealing

Cable penetration

Cable sealant

Fire stop

Flame resistant

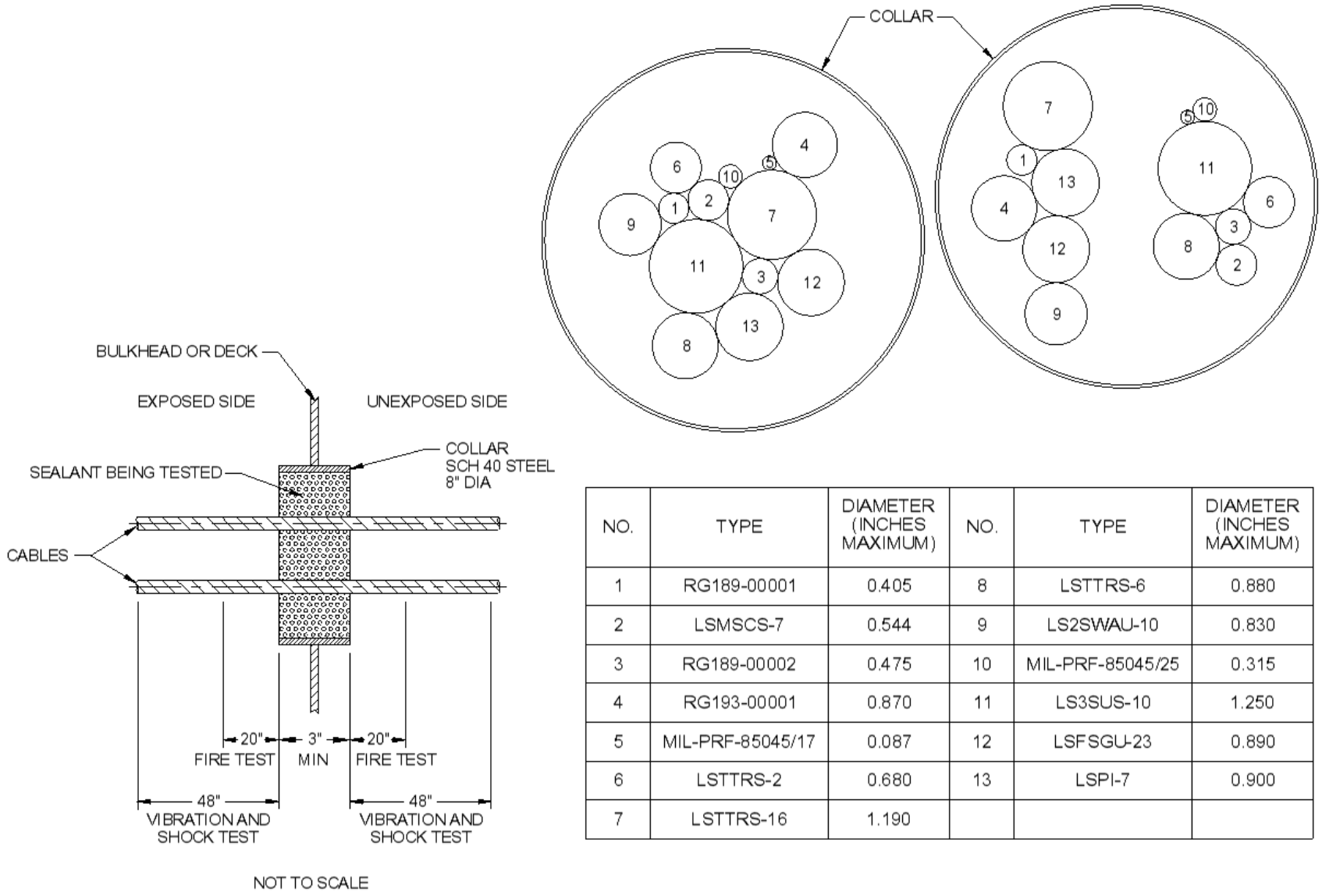
Heat expansion

Heat resistant

Stuffing tube sealant

6.11 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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NO.	TYPE	DIAMETER (INCHES MAXIMUM)	NO.	TYPE	DIAMETER (INCHES MAXIMUM)
1	RG189-00001	0.405	8	LSTTRS-6	0.880
2	LSMSCS-7	0.544	9	LS2SWAU-10	0.830
3	RG189-00002	0.475	10	MIL-PRF-85045/25	0.315
4	RG193-00001	0.870	11	LS3SUS-10	1.250
5	MIL-PRF-85045/17	0.087	12	LSFSGU-23	0.890
6	LSTTRS-2	0.680	13	LSPI-7	0.900
7	LSTTRS-16	1.190			

FIGURE 1. Cable types and collar configurations.

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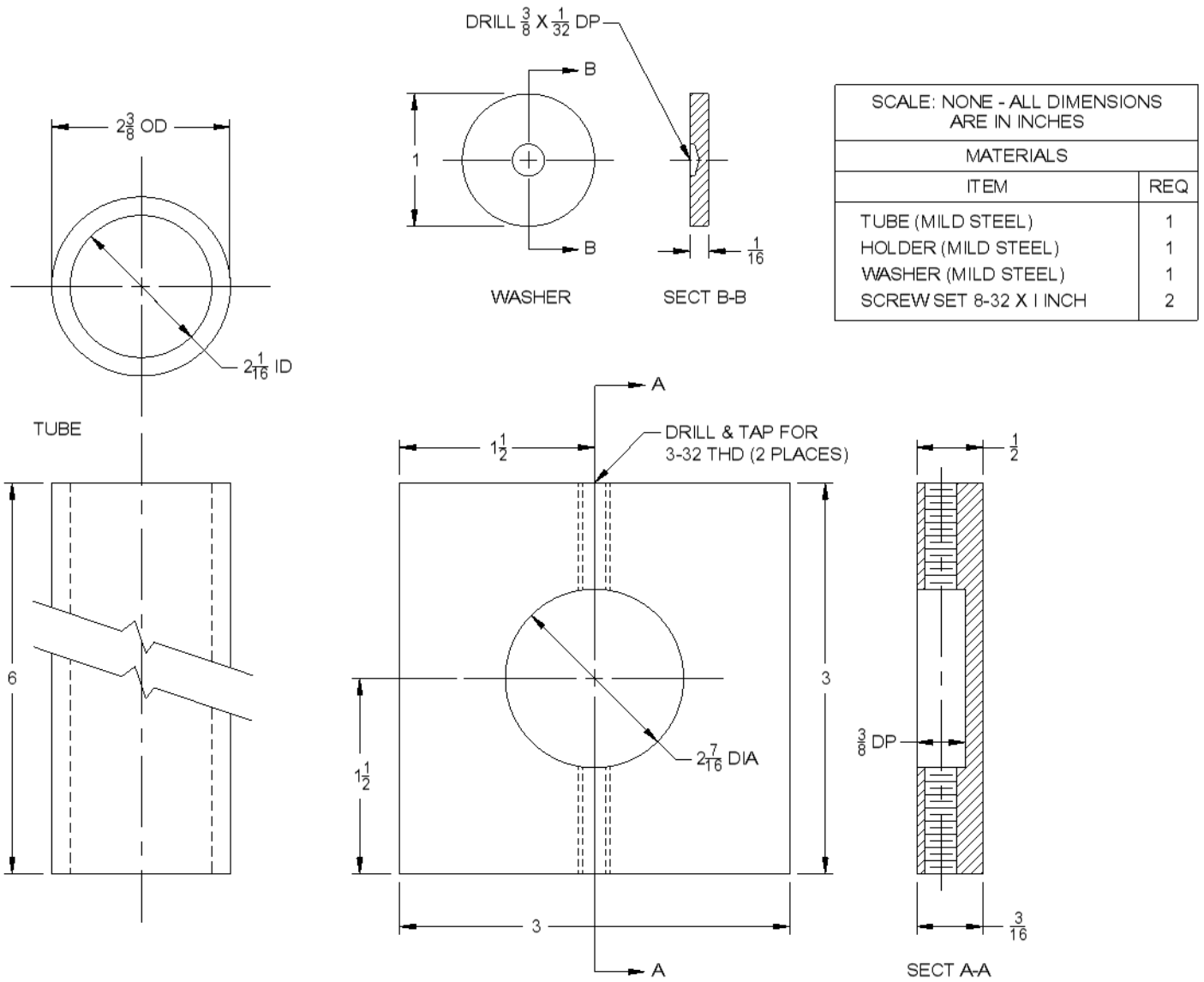


FIGURE 2. Tube holder.

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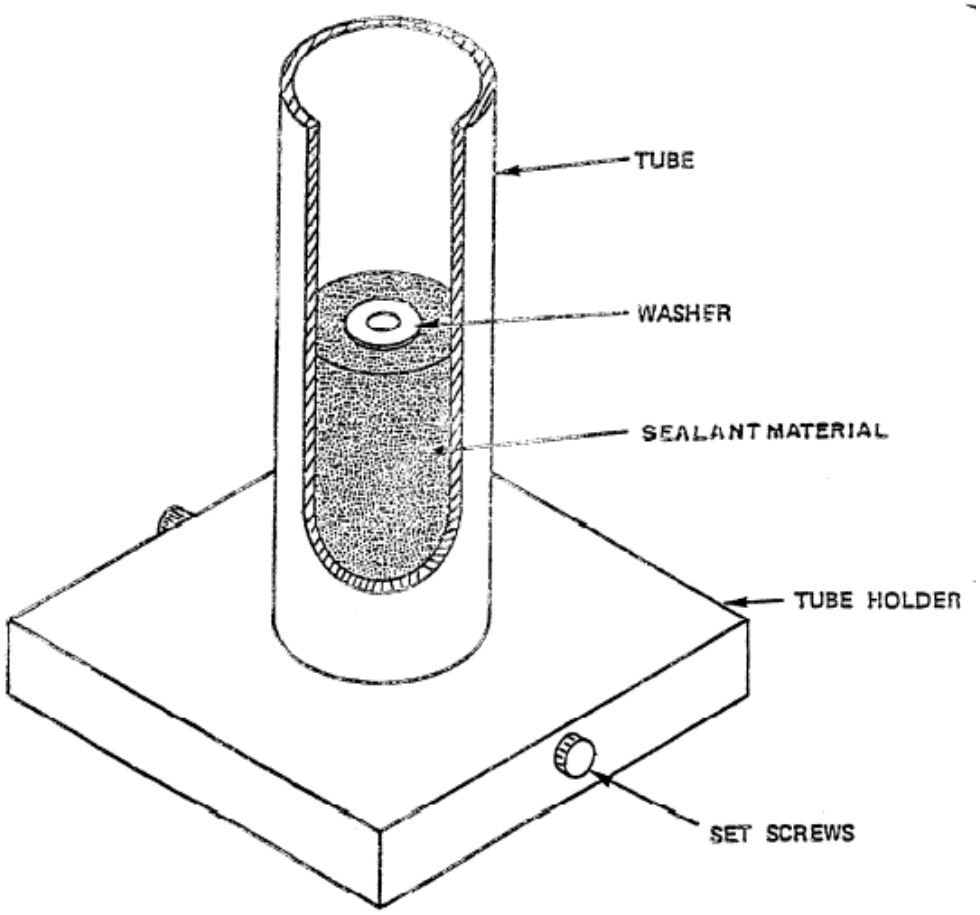


FIGURE 3. Assembly view of sample holder tube.

CONCLUDING MATERIAL

Custodians:
Army – CR
Navy – SH
Air Force – 85

Review activities:
Army – MI
DLA – GS

Preparing activity:
Navy – SH
(Project 5970-2024-002)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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